UWB WEBINAR SERIES

RESTRICTED

SESSION 1:
Introducing Ultra-Wideband Ranging

CAS HAMBURG
JULY 2022

SECURE CONNECTIONS FOR A SMARTER WORLD



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OVERVIEW ON PLANNED NXP WEBINARS ON UWB

JUL	Introducing Ultra-Wideband Ranging	Introducing Ultra-Wideband Radar
AUG	Introducing NCJ29D6	
SEP	Introducing NCJ29D6 Customer Demo Board	
OCT	Getting Started in SW development	UWB Radar with NCJ29D6
	...	

The webinar recording and slides will be available on nxp.com (Secure Files) shortly after the presentation.

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CONTENT OVERVIEW

- Time of Flight ranging
- Ultra-Wideband pulse
- Multipath Propagation and Channel Impulse Response
- IEEE Frame Format
- UWB Receiver Concept
- Q&A

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Time of Flight ranging

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TWO WAY RANGING – SINGLE SIDED (TWR-SS)

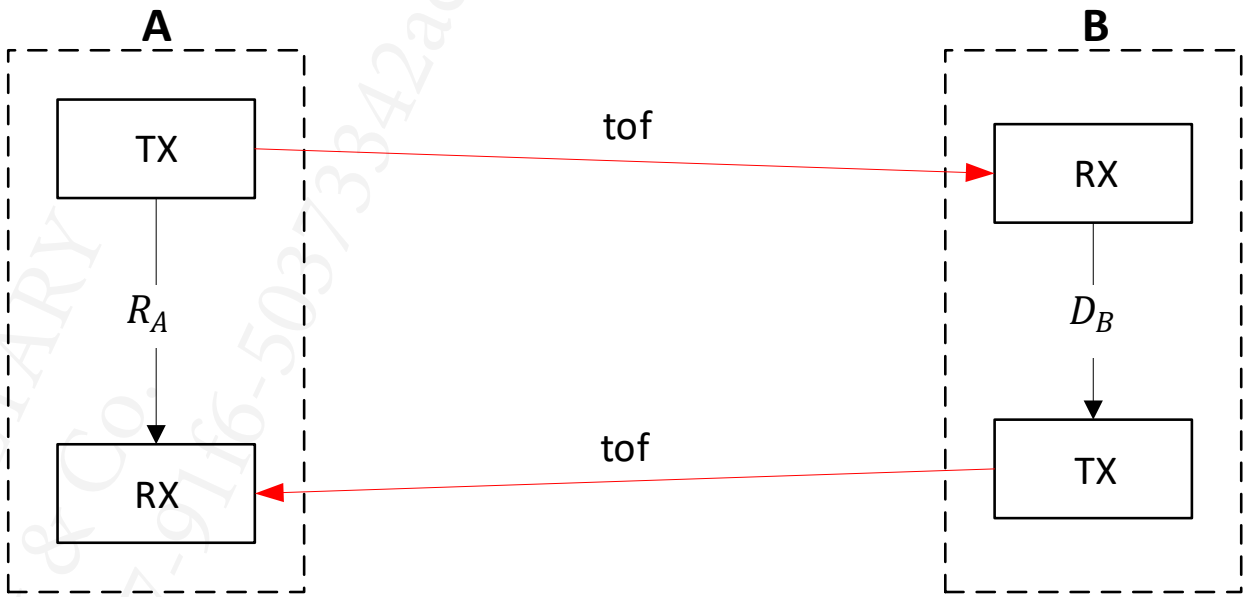
Consists of the following:

1. TX from device A (Poll)
2. RX on device B
3. TX from device B (Response)
4. RX on device A

Time between TX and RX on dev. A:
R_A = Round Time A

Time between RX and TX on dev. B:
D_B = Delay Time B

$$tof = \frac{R_A - D_B}{2}$$



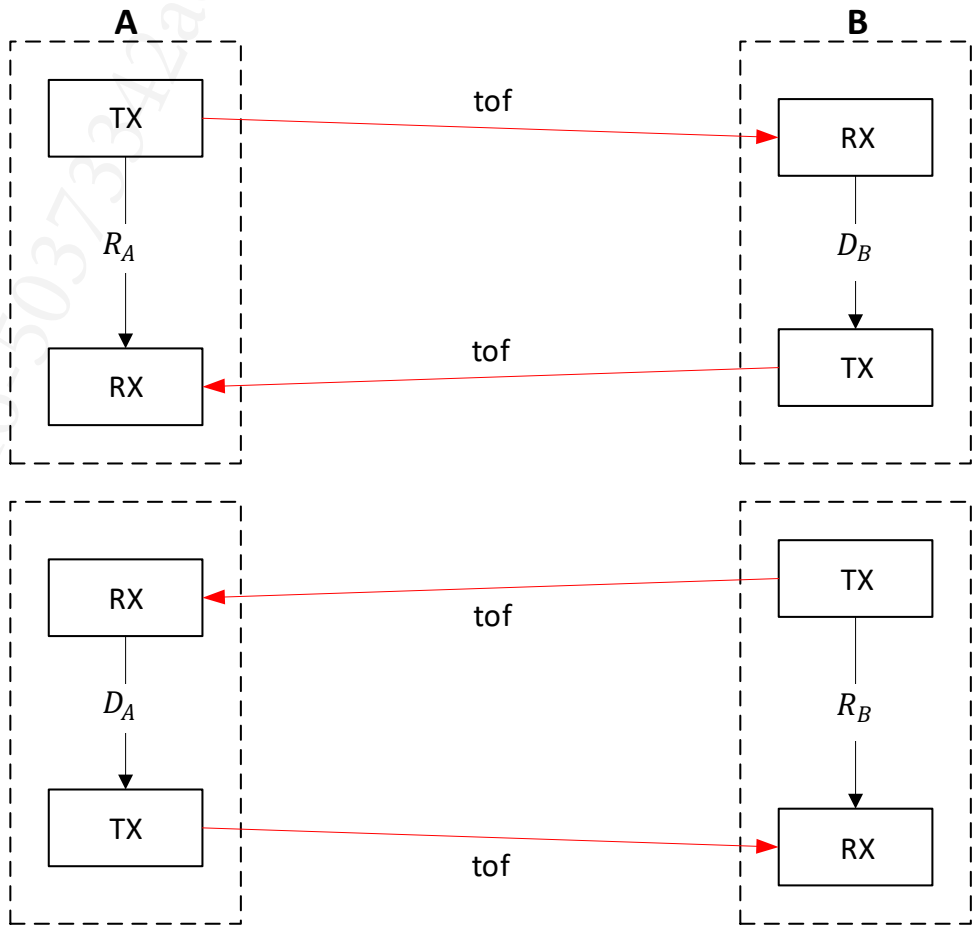
+ Simple
+ Low power consumption (only 2 frames)
– – Clock difference leads to big error

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TWO WAY RANGING – DOUBLE SIDED (TWR-DS)

To compensate the error introduced by the clock:

- Perform two TWR-SS
- One is started by device A, the other by device B
- The response message of the first TWR-SS and the poll message of the second TWR-SS can be combined (see next slide)



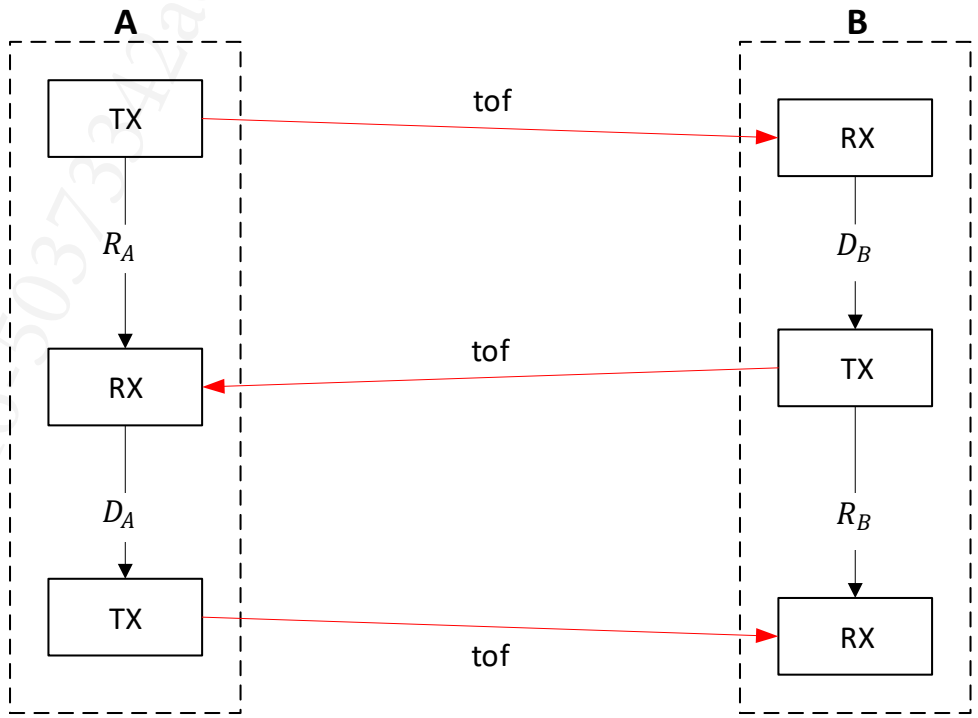
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TWO WAY RANGING – DOUBLE SIDED (TWR-DS)

Ranging consists of the following frames:

- Poll (A → B)
- Response (B → A)
- Final (A → B)

$$tof = \frac{R_A R_B - D_A D_B}{R_A + R_B + D_A + D_B}$$



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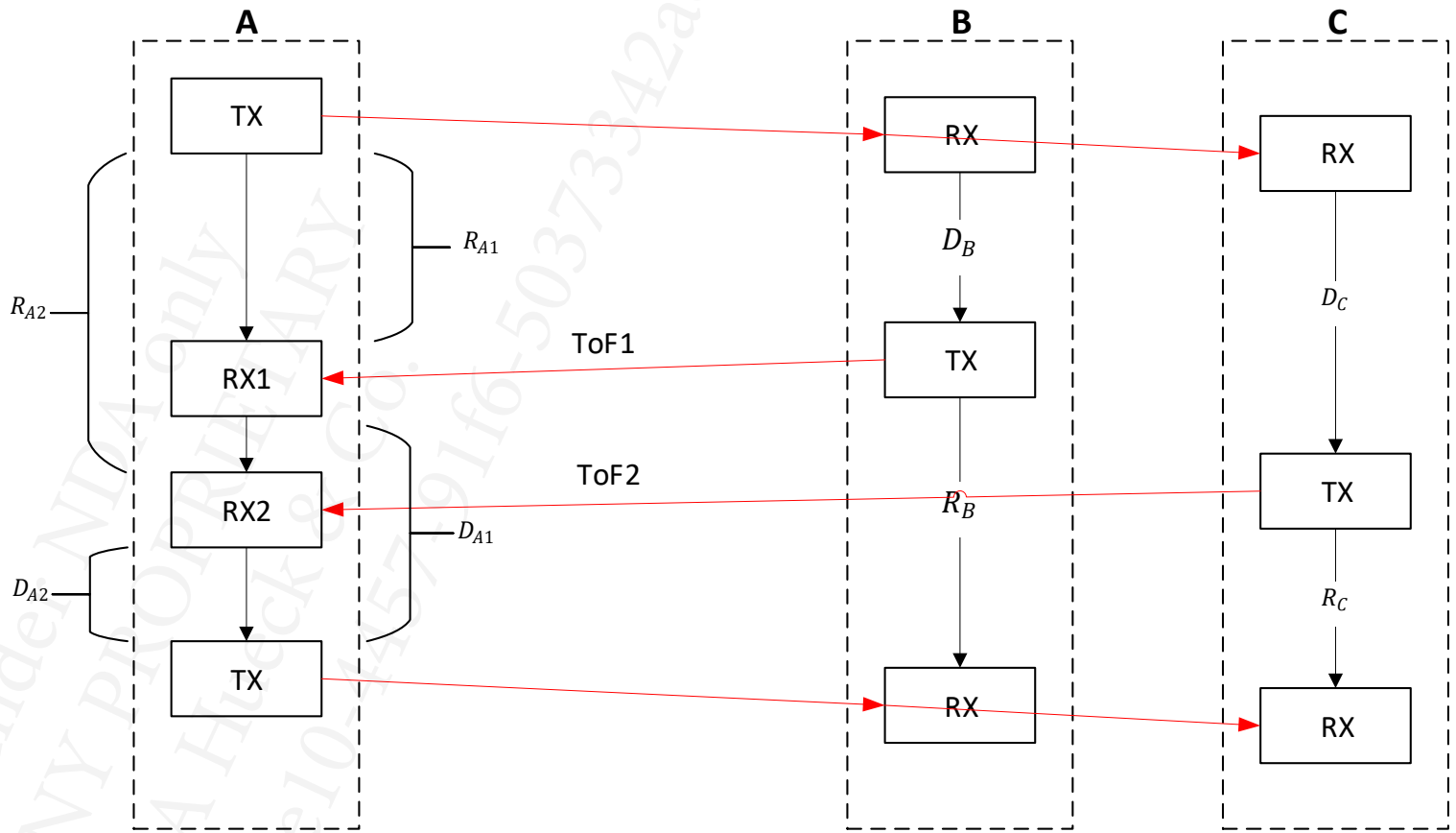
\+ Automatic compensation for clock differences in the two devices

– Requires more energy (3 messages)

INTERLEAVED TWO WAY RANGING – DOUBLE SIDED (TWR-DS)

$$ToF1 = \frac{R_{A1}R_B - D_{A1}D_B}{R_{A1} + R_B + D_{A1} + D_B}$$

$$ToF2 = \frac{R_{A2}R_C - D_{A2}D_C}{R_{A2} + R_C + D_{A2} + D_C}$$



Example with 1 initiator A (mobile phone) and two anchors B and C.

Can easily be extended to multiple anchors.

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Ultra Wideband pulse

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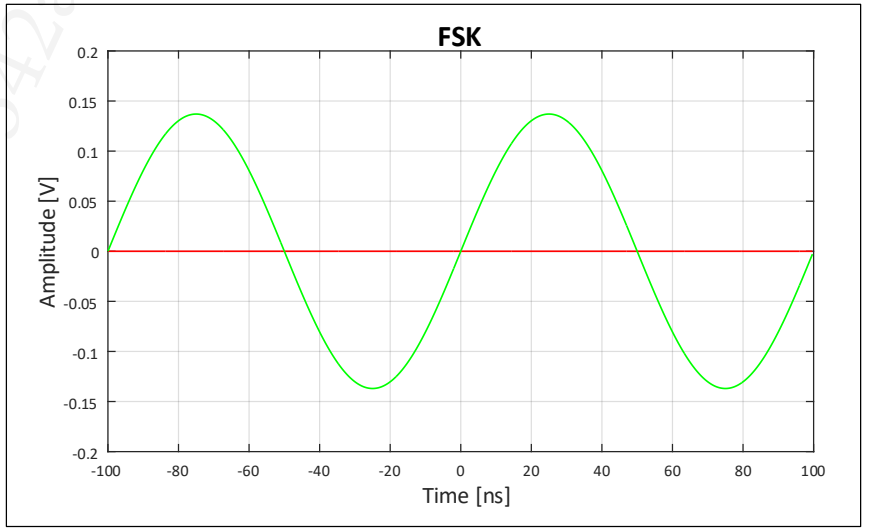
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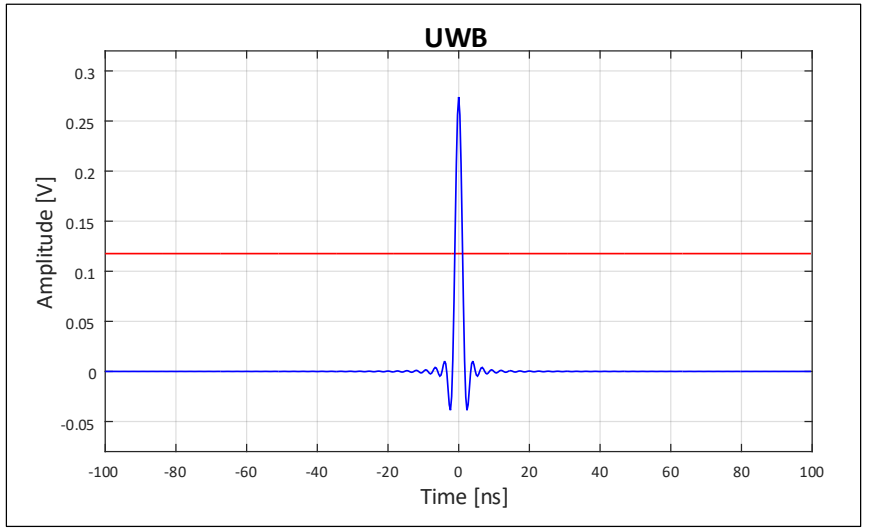
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DIFFERENCE BETWEEN UWB AND NARROWBAND

- Top: Narrow Band signal in time domain
  - Threshold crossing is periodic



- Bottom: Ultra Wide Band signal in time domain
  + Unique point in time for the pulse crossing the threshold

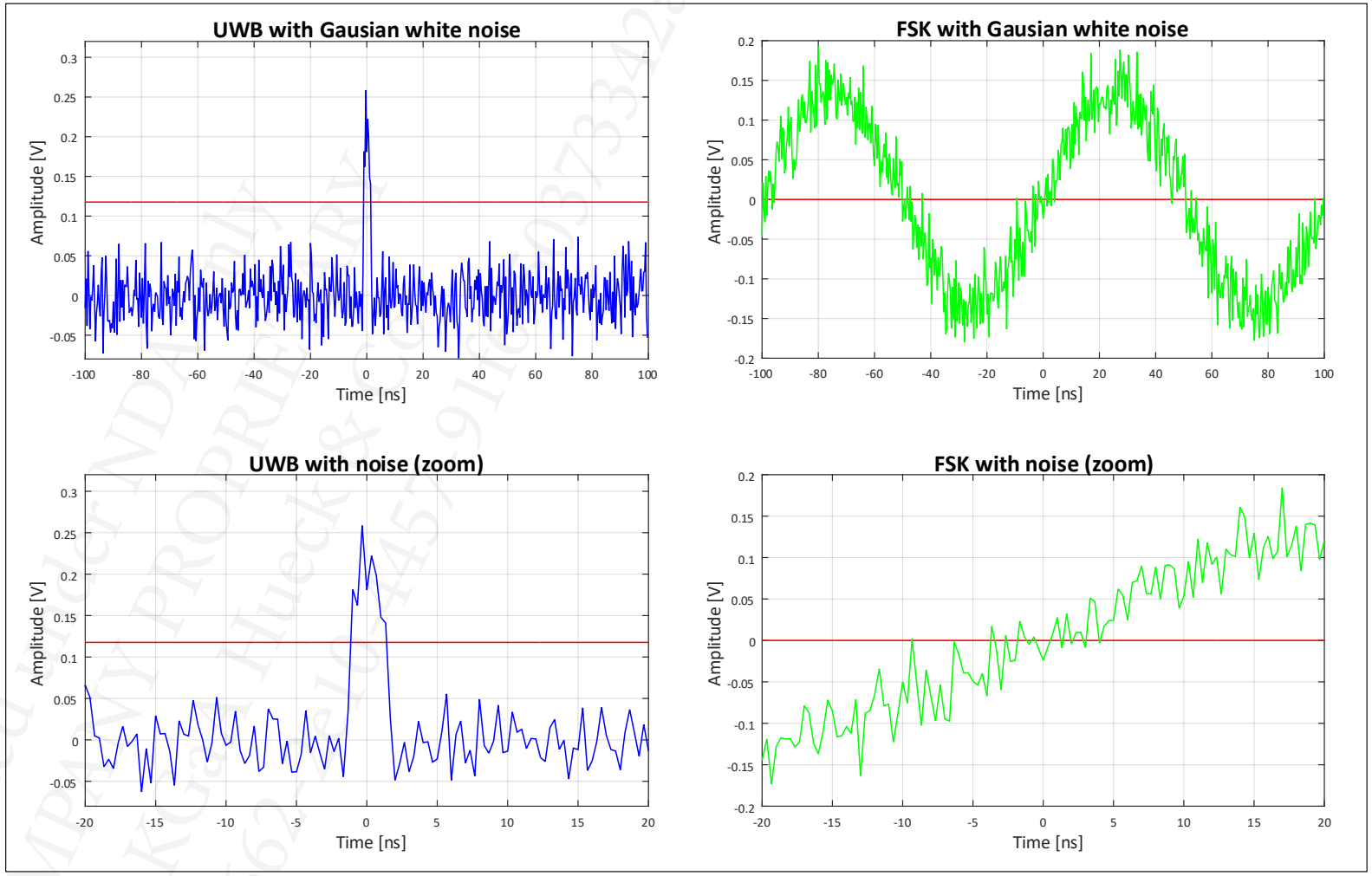


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DIFFERENCE BETWEEN UWB AND NARROWBAND

- Signals + Noise
- UWB: Point in time still unique
- NB: Not unique anymore



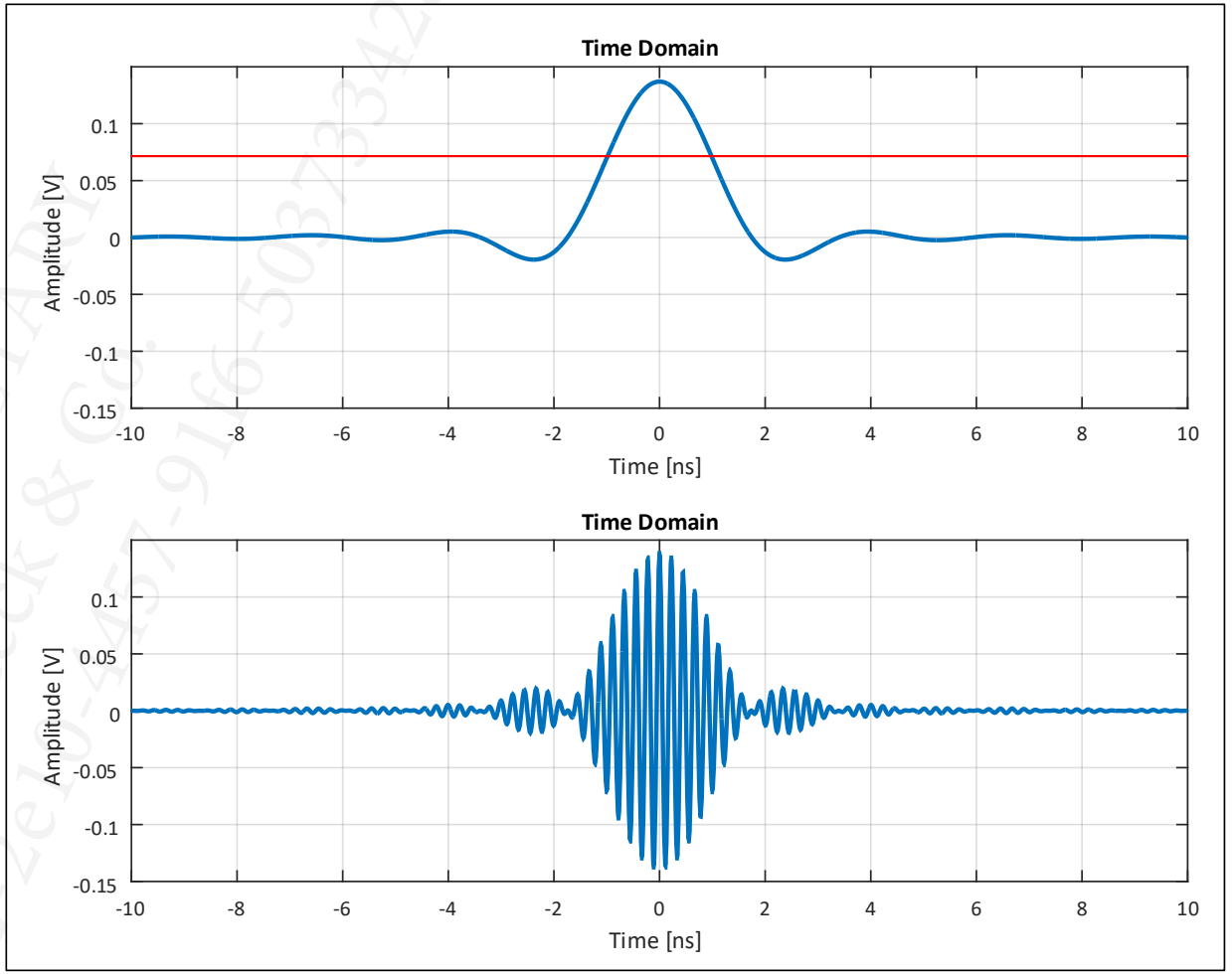
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UWB PULSE SHAPE

- UWB Pulse:
  - Tp =~2 ns
  - 3 dB bandwidth: $\frac{1}{2\ ns} = 500\ MHz$
  - Reference Pulse specified in IEEE

- Reference Pulse:

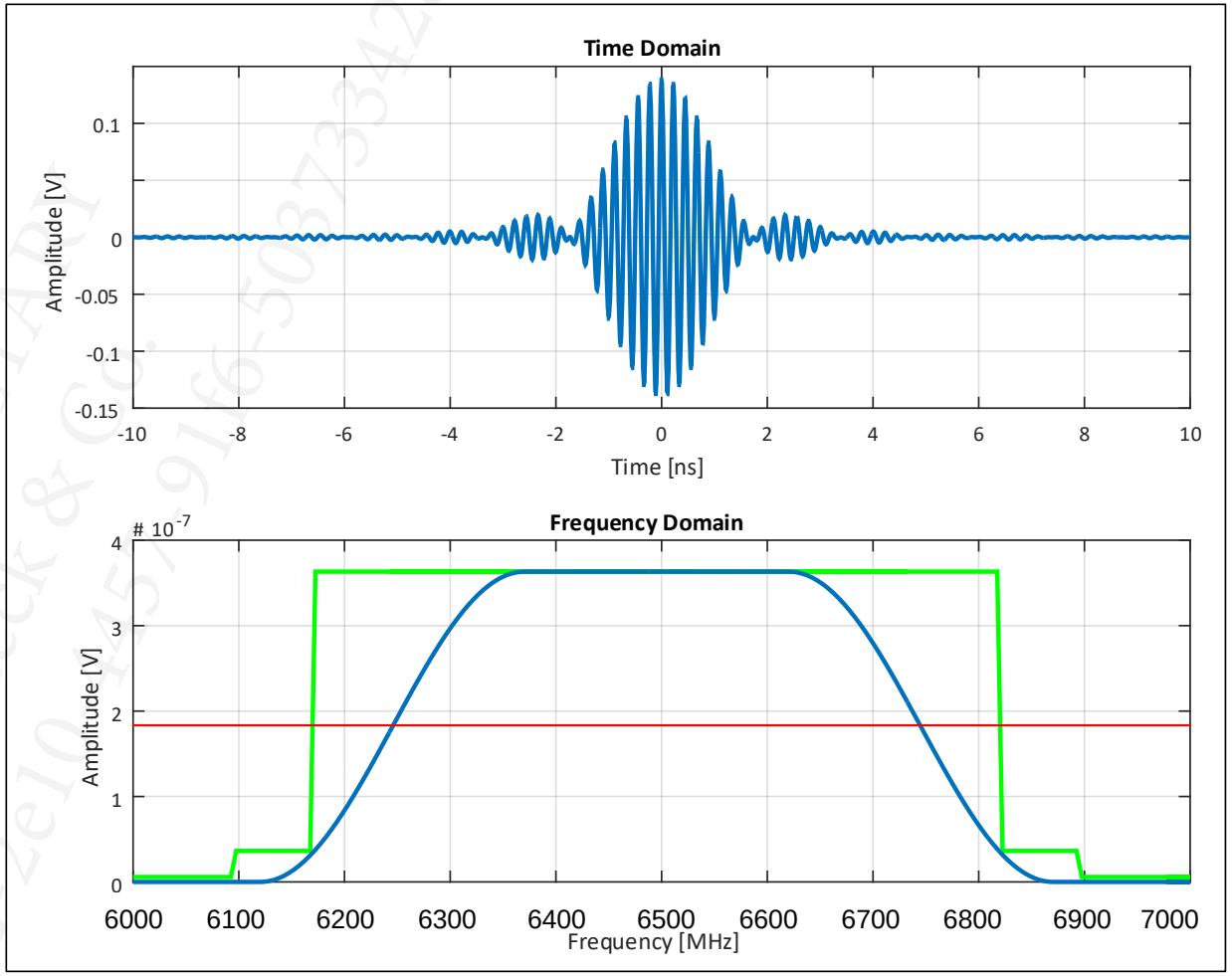
$$r(t) = \frac{4\beta}{\pi\sqrt{T_p}} \cdot \frac{\cos\left(\frac{(1+\beta)\pi t}{T_p}\right) + \frac{\sin\left(\frac{(1-\beta)\pi t}{T_p}\right)}{4\beta\left(\frac{t}{T_p}\right)}}{1 - \left(4\beta\frac{t}{T_p}\right)^2}$$



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UWB PULSE SHAPE

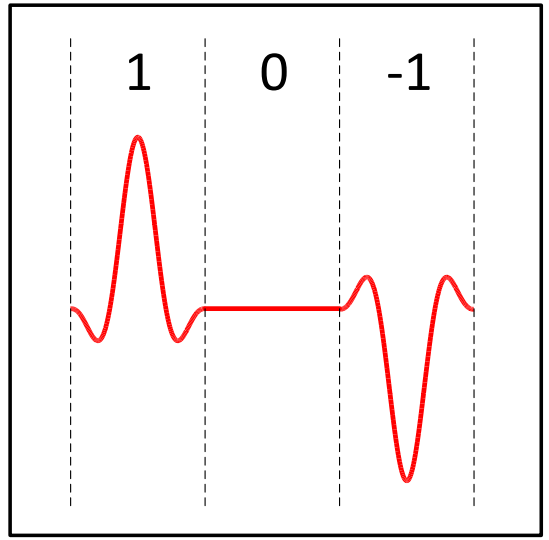
- UWB pulse is mixed with carrier frequency. Carrier frequency defines the channels:
  - 5: 6489.6 MHz
  - 6: 6988.8 MHz
  - 8: 7488.0 MHz
  - 9: 7987.2 MHz
- 3 dB Bandwidth: $\frac{1}{2\ ns} = 500\ MHz$



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TERNARY CODE

- With UWB pulses, 3 states are coded (ternary code):
  - Phase is used to modulate +1 or -1
  - No pulse is interpreted as a 0



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Multipath Propagation and Channel Impulse Response



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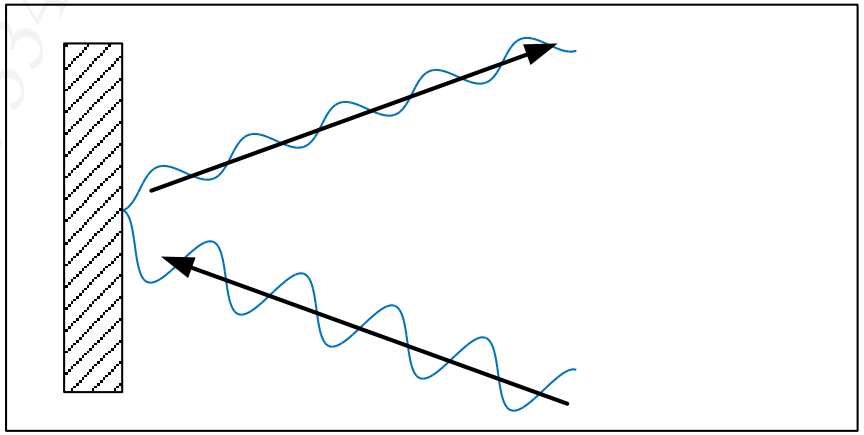
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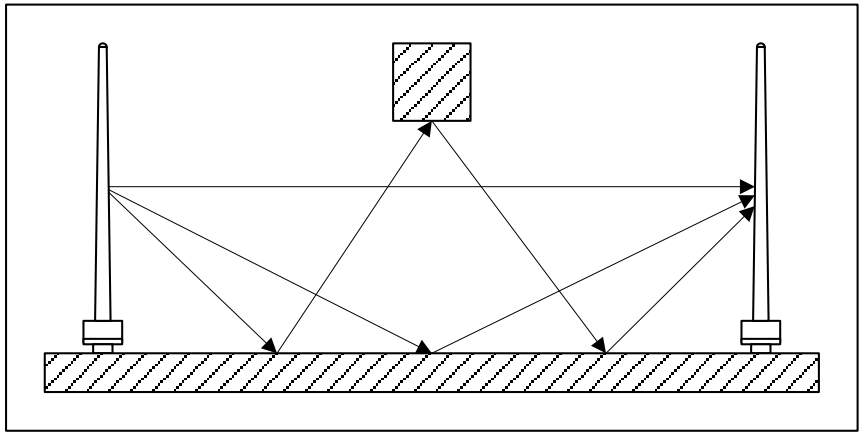
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MULTIPATH ENVIRONMENT

- Radio waves are being reflected on objects
- Attenuation and phase shifting occurs

- The same message will be received multiple times because of different paths with different lengths



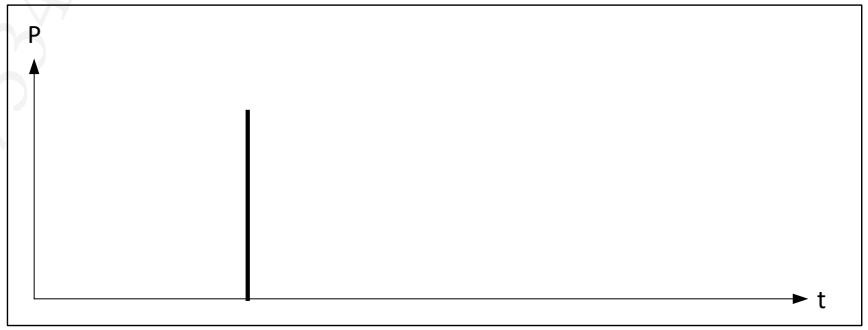


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16.

CHANNEL IMPULSE RESPONSE - 1

- The CIR shows the received power over time
- One path:
  - Only one discrete value



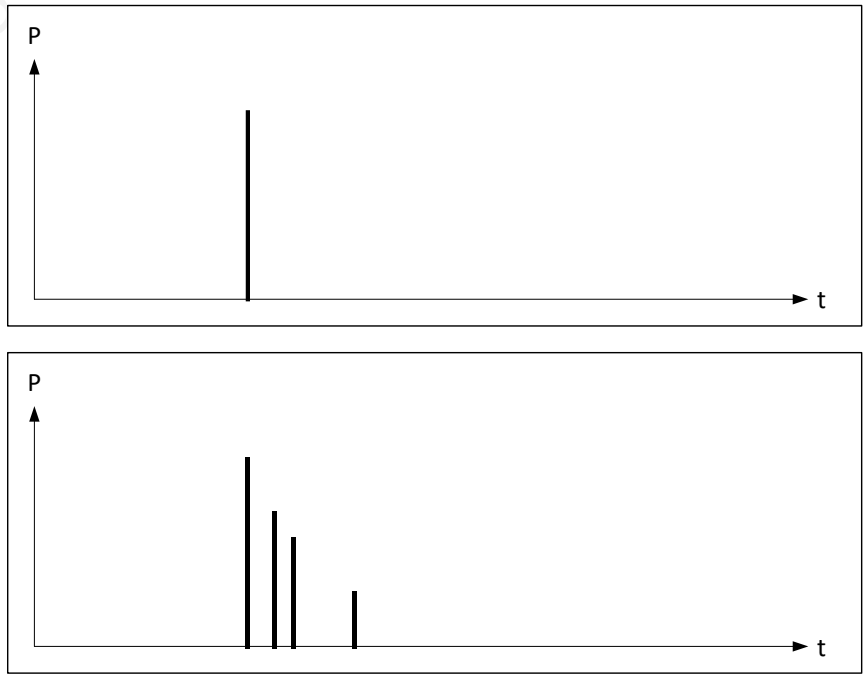
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17.

CHANNEL IMPULSE RESPONSE - 2

- The CIR shows the received power over time
- One path:
  - Only one discrete value

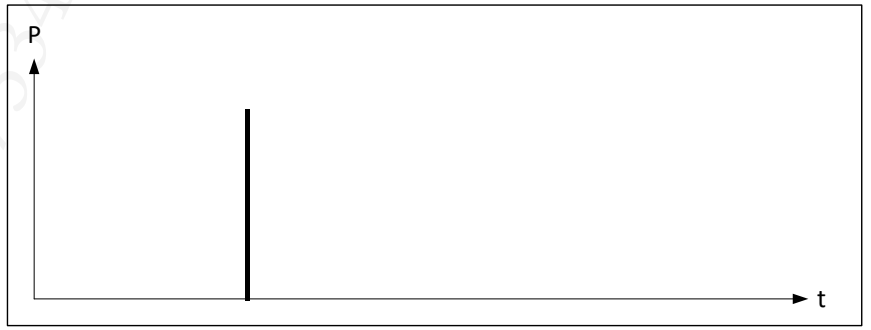
- Multi-Path: Line of Sight (LoS):
  - Multiple paths, the first one is the strongest



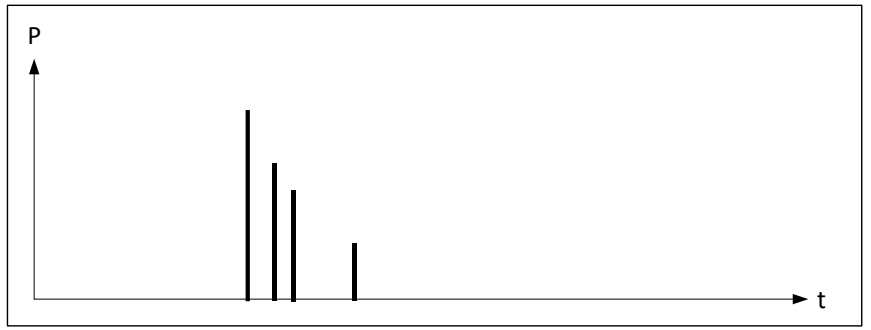
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CHANNEL IMPULSE RESPONSE - 3

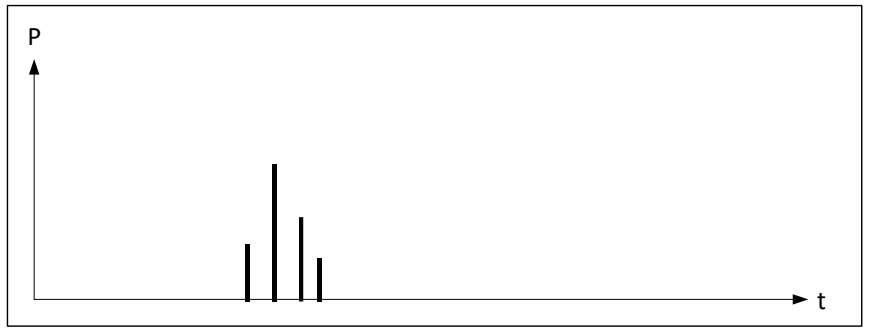
- The CIR shows the received power over time
- One path:
  - Only one discrete value



- Multi-Path: Line of Sight (LoS):
  - Multiple paths, the first one is the strongest



- Multi-Path: None Line of Sight (nLoS):
  - Multiple paths, the first one may not be the strongest



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IEEE Frame Format



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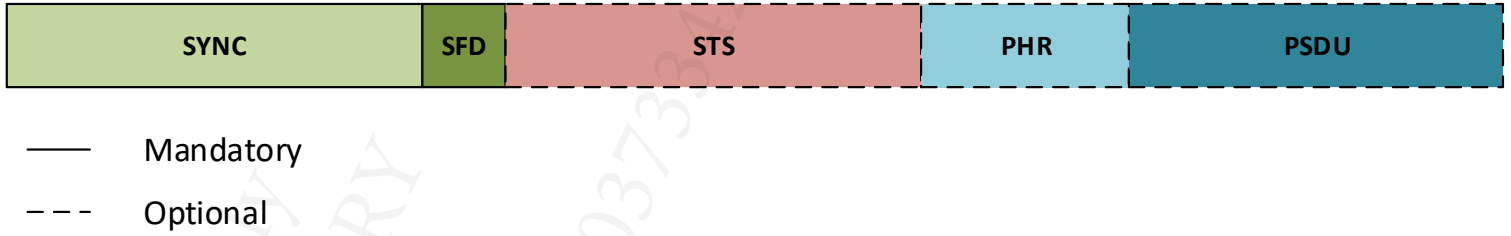


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UWB FRAME

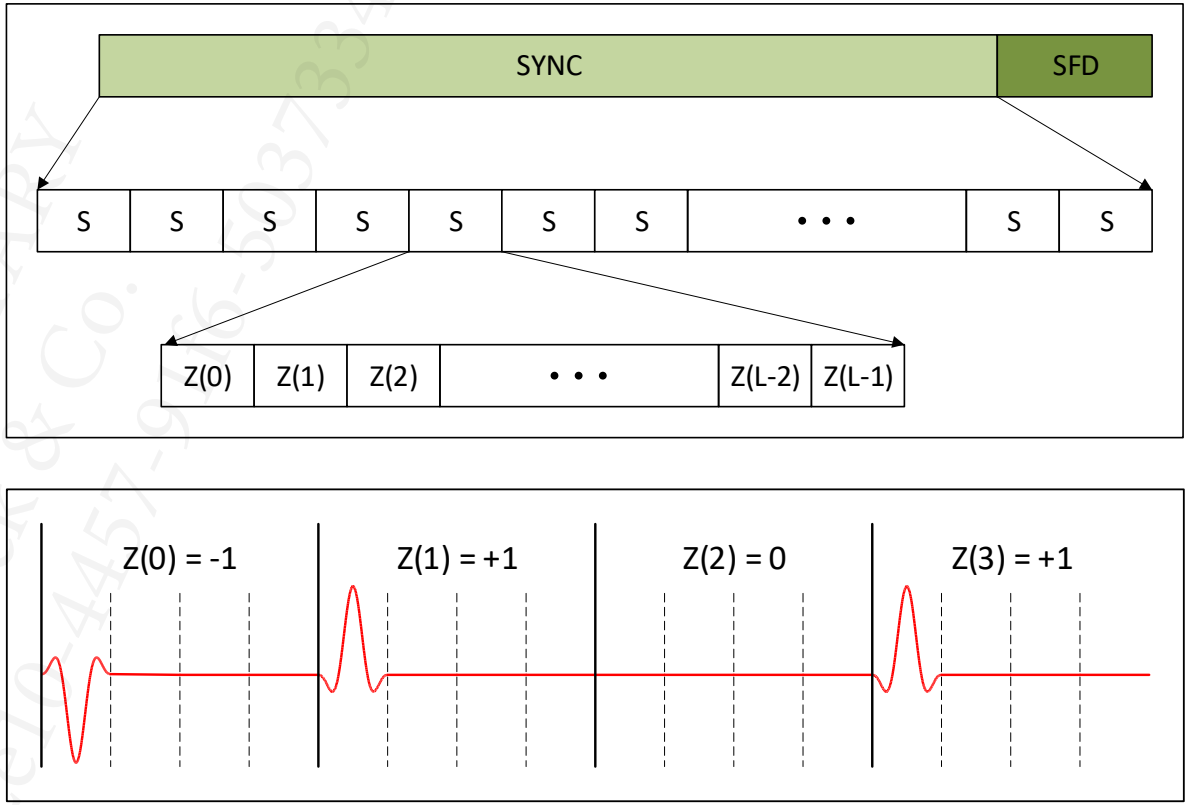


- SYNC:
  - Preamble
  - For synchronization
- SFD – Start Frame Delimiter:
  - Break in preamble
  - Time Stamp Reference Marker
- STS – Scrambled Timestamp Sequence
  - Encrypted sync field
- PHR – PHY Header
  - Header information regarding the payload
- PSDU – PHY Service Data Unit
  - Payload

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SYNC FIELD

- Sync field consist of N preamble symbols *S*
- Each symbol *S* has L ternary characters *Z*
- Each *Z* is either +1, -1 or 0.
- Each *Z* has multiple 2ns long slots. Final spreading (i.e. [1,0,0,0]) increases distance between the pulses and thus reduces pulse repetition frequency.



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SYNC CODES

Sync codes are designed to show optimal autocorrelation properties, e.g., minimal sidelobes during correlation.

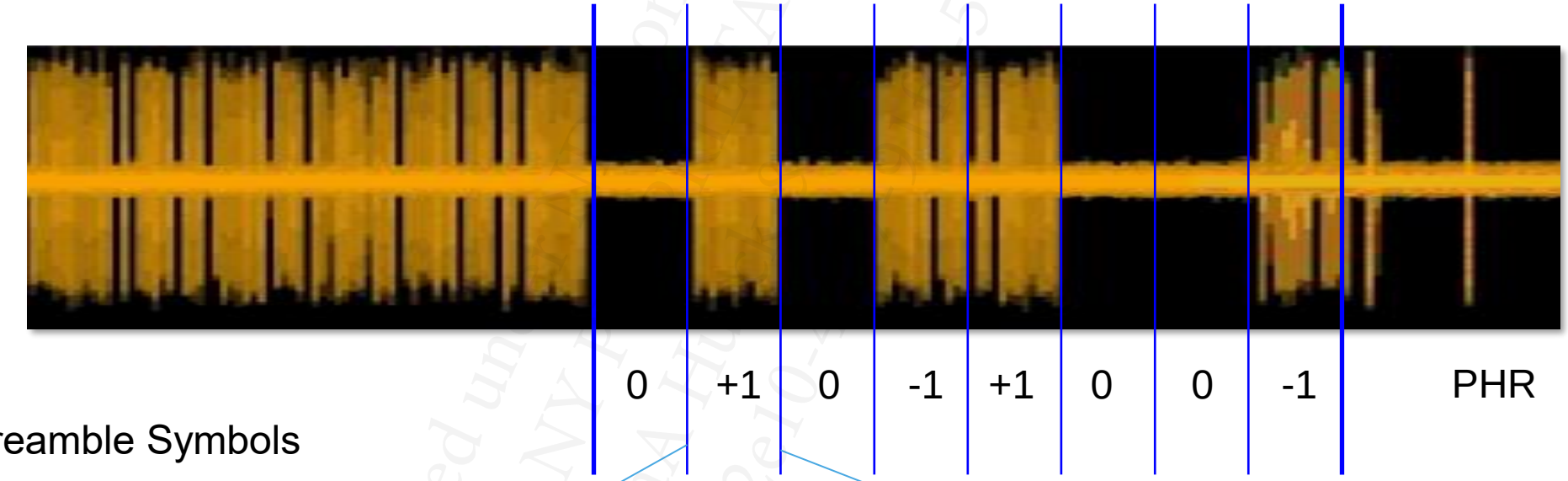
This property enables the receiver to separate the signal from the background noise and other disturbers.

- Code Nr 1 for PRF16:
  - -0000+0-0+++0+-000+-+++00-+0-00
- Code Nr 9 for PRF64:
  - +00+000-0--00--+0+0+00-+-++0+0000++-000+00-00--0-+0+0--0-+++0++000+-0+00-0++-0+++00-+00+0+0-0++-+--+000000+00000-+0000-0-000--+
- Code Nr 1 for PRF128:
  - -+++++++-+-+-+--++--+++-+++-+--+-++---++-++++-++-+-++-+++--+--+---+++----+-++++++--+-+-+++--++-+---+--++++---+-+----++-----+-----

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SFD FIELD

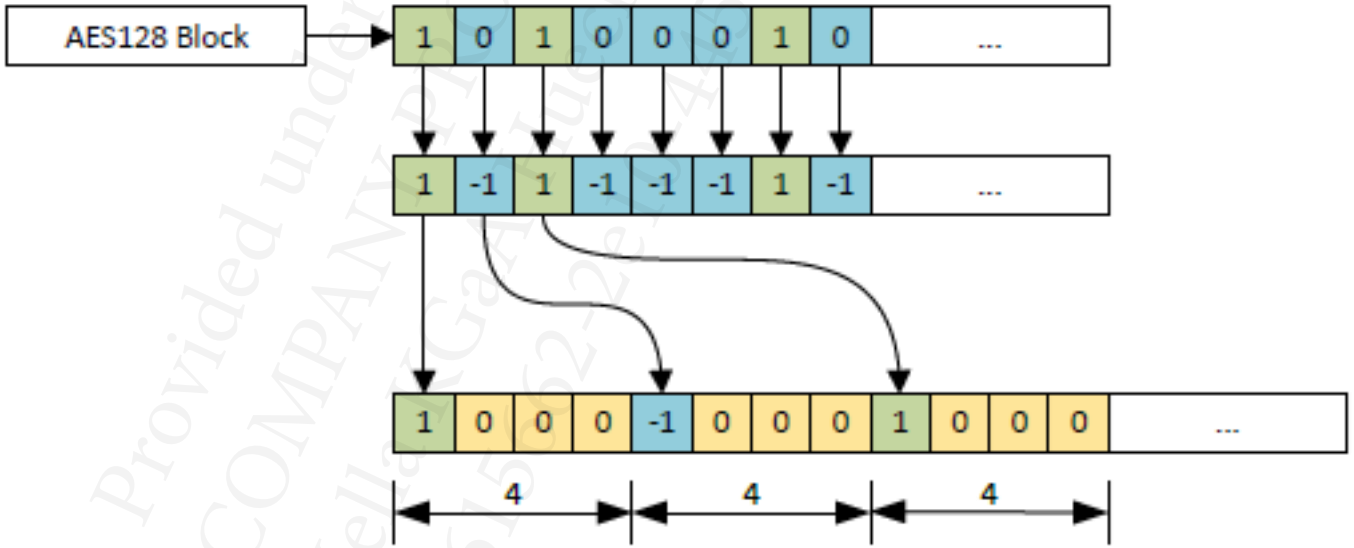
- SFD (Start Frame Delimiter) consists of the same preamble symbols as used in the Sync field, but is spread with the short SFD pattern defined in IEEE Std. 802.15.4: “0 +1 0 -1 +1 0 0 -1”



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STS (SCRAMBLED TIMESTAMP SEQUENCE)

- The STS has been added to IEEE 802.15.4z standard for secure ranging. Encoding is similar to the SYNC field.
- It is a secret number, generated by cryptographic algorithms (AES128).
- The same STS must be generated on both communicating devices.
  - No frame reception with non-synchronized STS!
- Encoding is similar to the SYNC field (spreading factor, binary coding)



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Ultra-Wideband Receiver Concept



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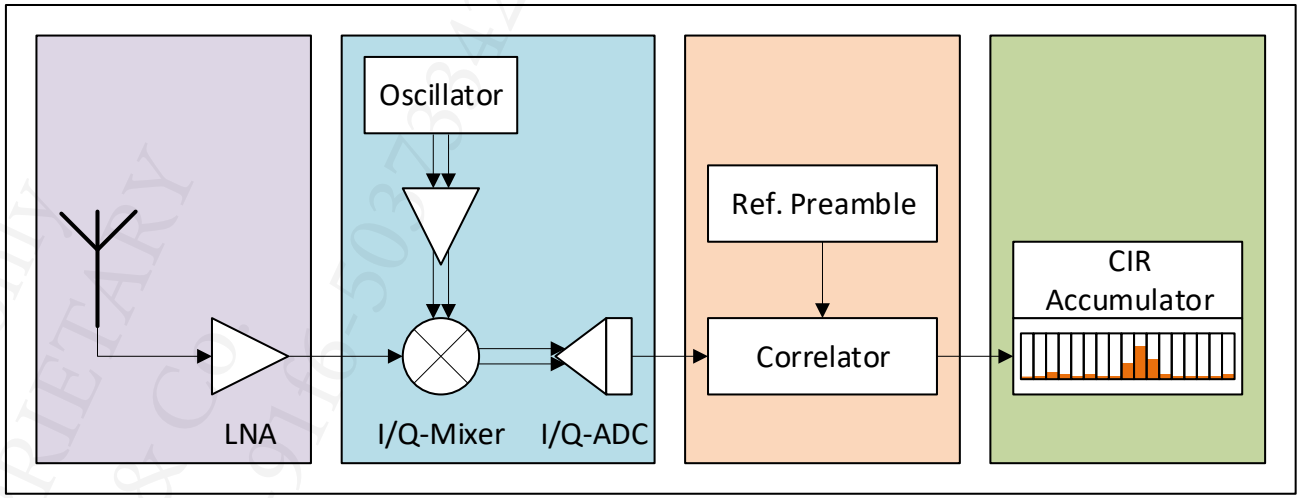
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RECEIVER

- LNA for amplifying the incoming signal

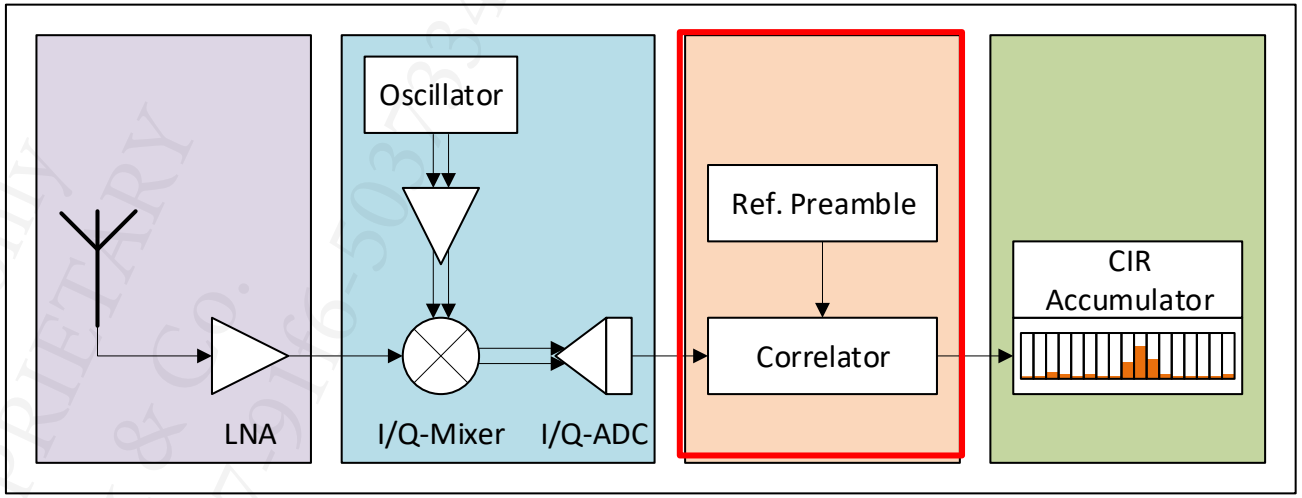


- I/Q mixer and ADC
  - Mix signal down to baseband
  - Convert to digital
- Correlator
  - Correlates the received signal with the reference preamble symbols
  - Used as a de-spreader to recombine the frame pulses to get the channel impulse response
- CIR Accumulator
  - Accumulates the output of the correlator over the length of a SYNC symbol
  - Used for noise reduction

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CORRELATOR

- Takes the data stream from the I/Q-ADC and correlates it with a reference preamble
- The reference preamble must be the same preamble sequence that was also used on the transmitter site.

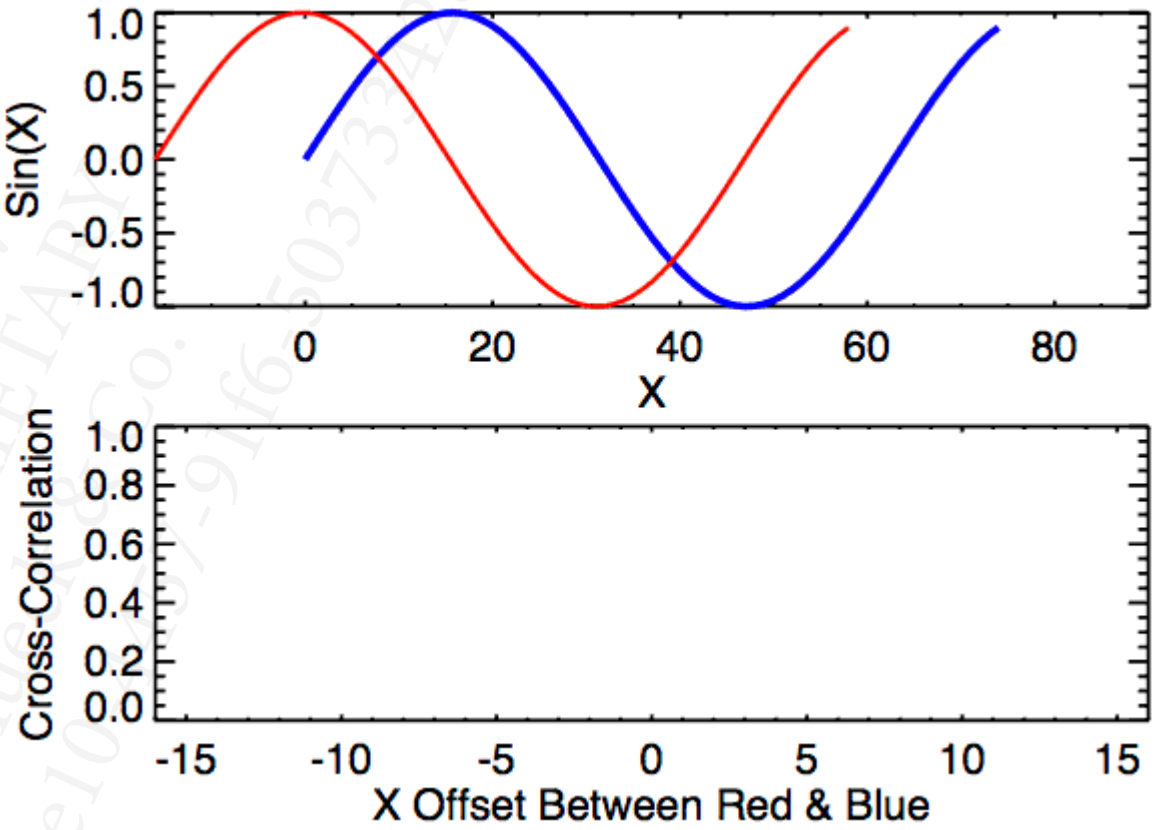


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EXEMPLARY CORRELATION

It demonstrates how two shifted (sinusoidal) signals create the correlation signal

The green signal is like our CIR:
1 = RED and BLUE are overlapping
0 = No correlation
-1 = (not shown) signals are inverse



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CORRELATION

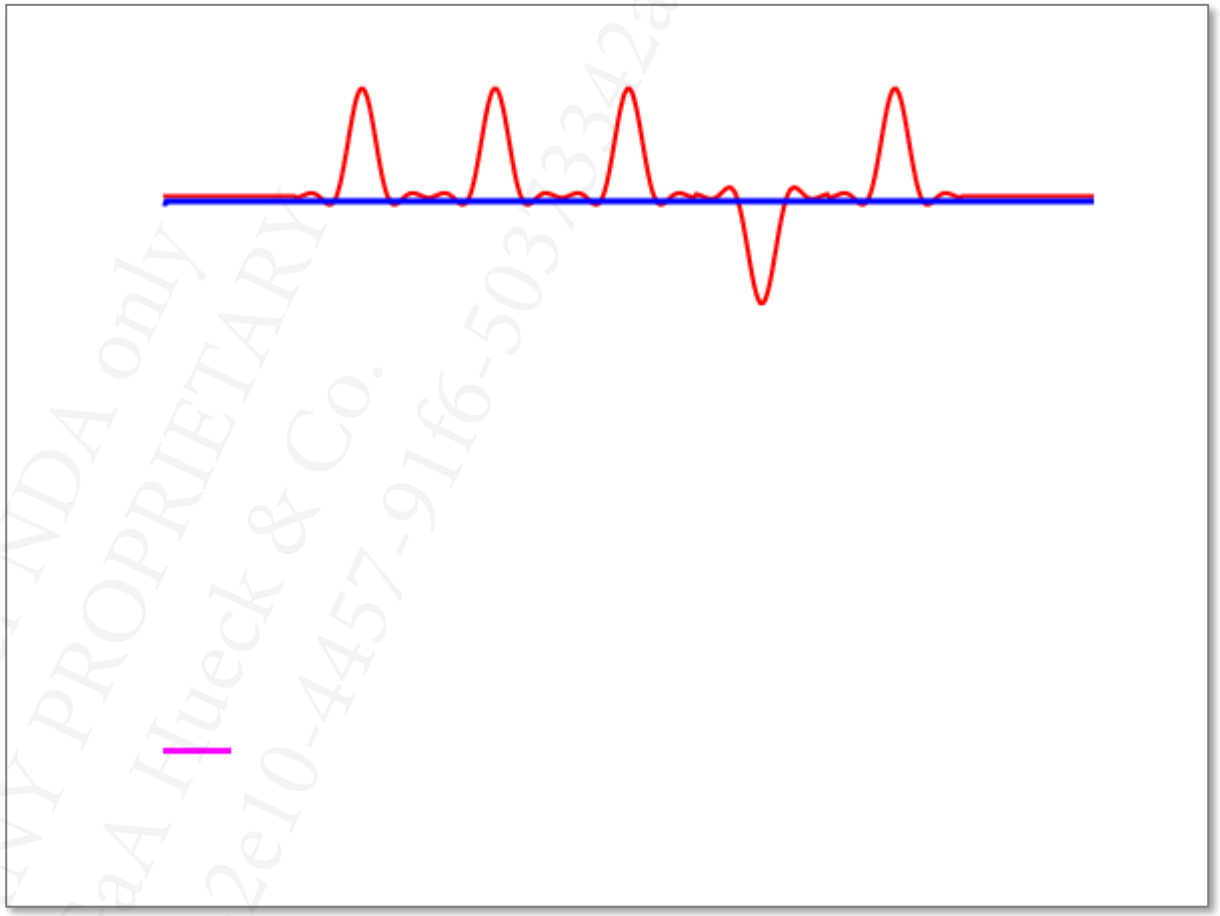
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Slide Show !!

It demonstrates how the UWB correlator is working.

Red = Received signal
Blue = Reference signal
Magenta = Correlation result

The maximum is given if both curves are congruent.

The small peaks shows the side lobes. In this case with a code length of 5, the side lobes have an amplitude of 1/5.



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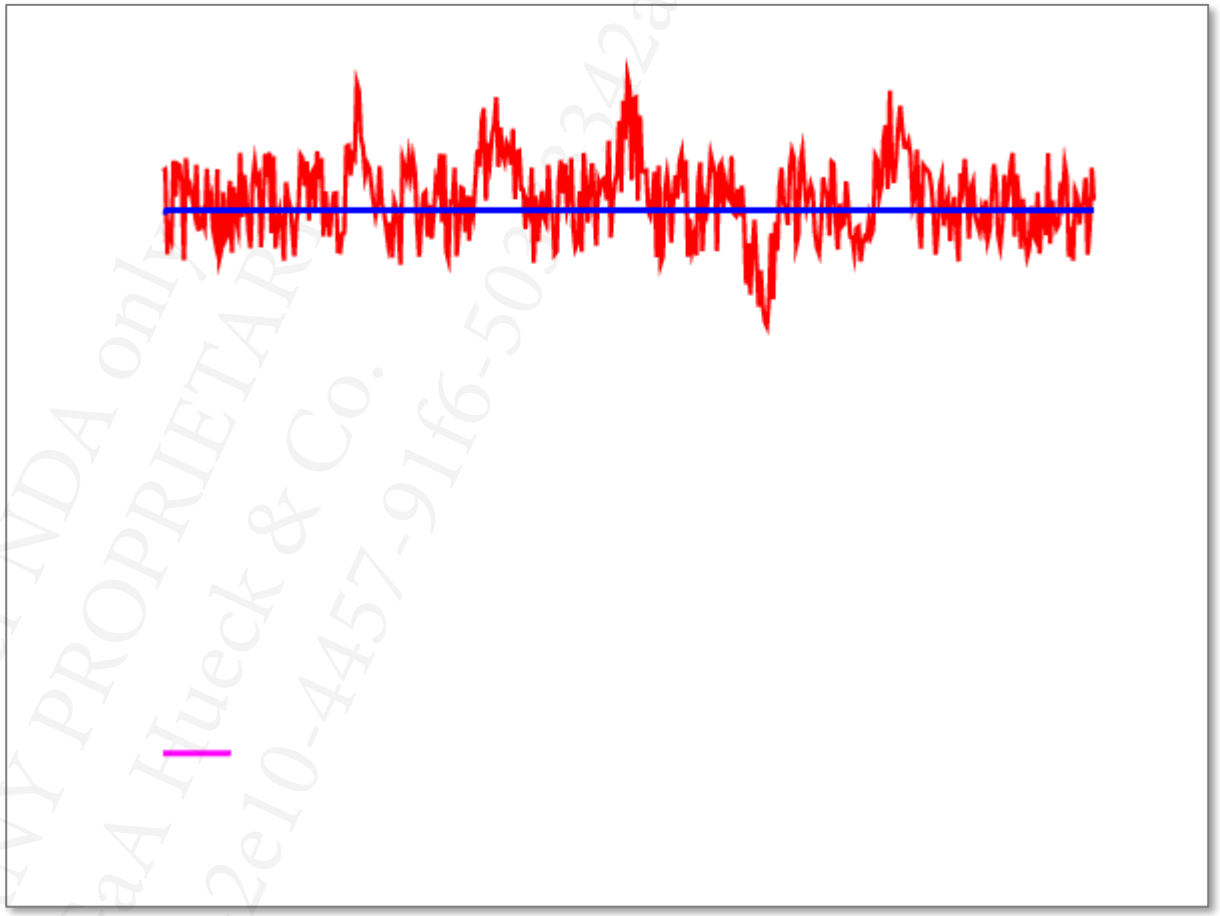
CORRELATION WITH NOISE

It demonstrates how the UWB correlator is working with noise.

Red = Received signal
Blue = Reference signal
Magenta = Correlation result

The maximum is given if both curves are congruent.

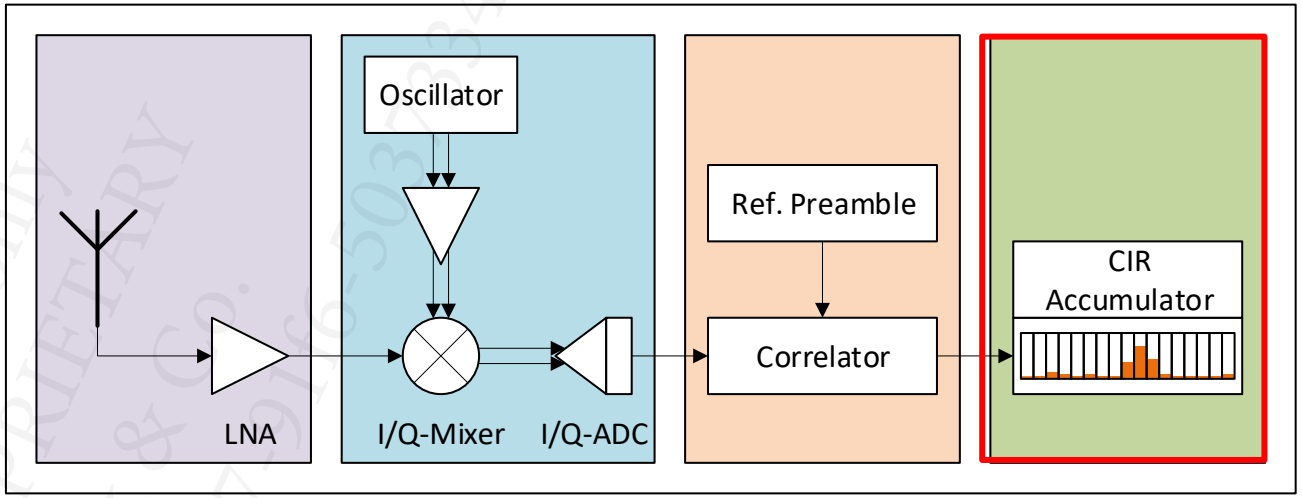
The small peaks shows the side lobes. In this case with a code length of 5, the side lobes have an amplitude of 1/5.



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ACCUMULATOR

- The accumulator takes the output of the correlator and sums up the results over time.
- Improves Signal to Noise Ratio (SNR)



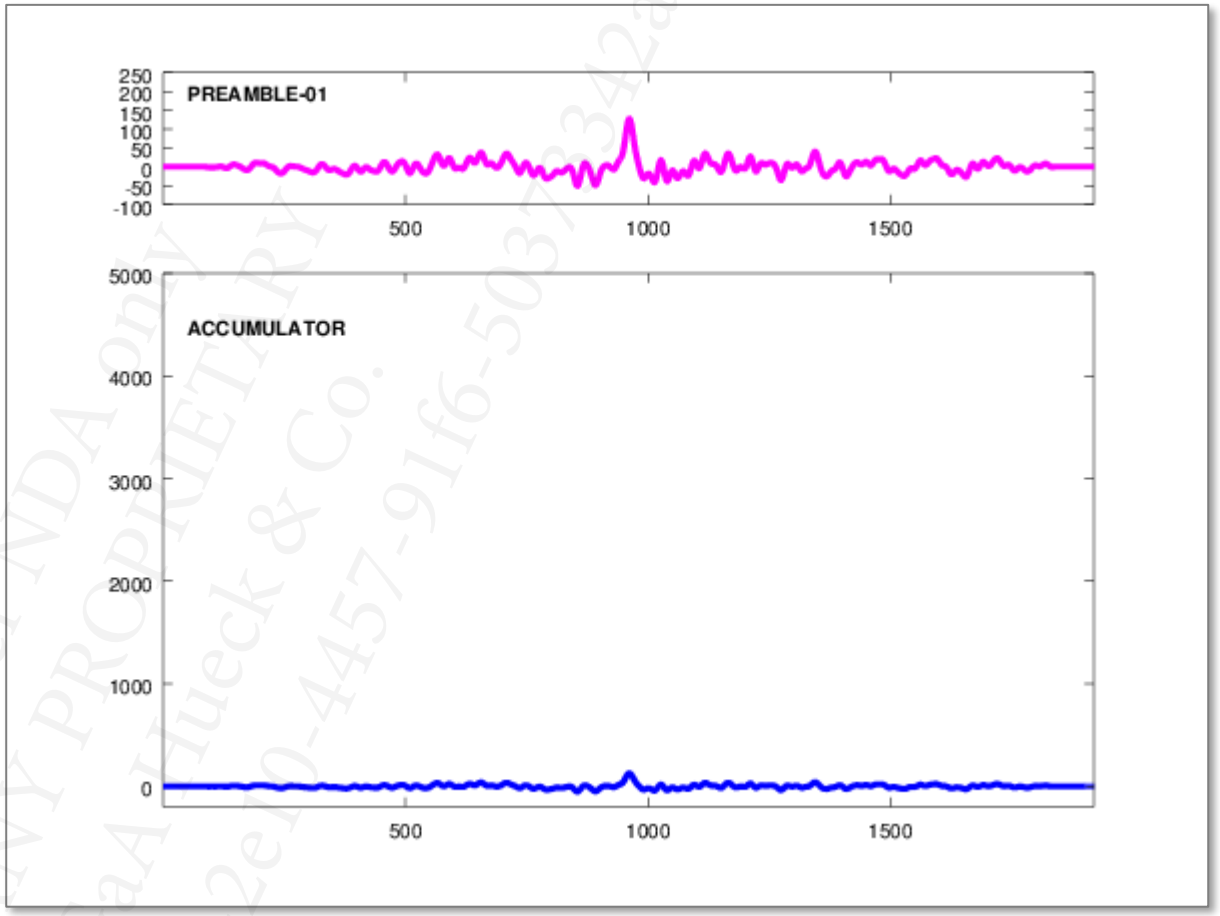
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ACCUMULATOR

It demonstrates how the UWB accumulator is working.

Magenta = Correlation with noise
Blue = Accumulator

It can be seen that the accumulator improves the signal to noise ratio with every symbol received.



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Q&A

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